

Dual-Innovation Neural-Kalman Fusion for Causal Indoor Localization

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Abstract—Indoor localization can combine Wi-Fi, magnetic-field, and inertial sensing, but practical fusion requires a causal mechanism for reconciling sparse spatial measurements with drifting pedestrian dead reckoning (PDR) when measurement availability and reliability vary over time. We propose a decoupled learned state-space estimator in which a Wi-Fi heatmap Multi-Layer Perceptron (MLP) and an 84-frame gravity-referenced magnetic Convolutional Neural Network (CNN) independently produce Cartesian position measurements, while PDR supplies relative motion. A DualKalmanNet forms separate Wi-Fi and magnetic innovations and uses a Gated Recurrent Unit (GRU) to predict independent 2×2 correction gains. Availability masks remove missing modalities, and a training-normalized magnetic log-uncertainty score scales the magnetic correction. Temporal fusion is evaluated on 60 independent map-constrained test trajectories within a fixed surveyed environment. With 1 Hz Wi-Fi, the uncertainty-weighted model obtains 0.494 m mean error (0.437 m median), compared with 0.473 m (0.449 m median) for Wi-Fi-only KalmanNet. Under degraded Wi-Fi (5 s updates with 40% AP dropout), it reduces mean error from 1.533 m to 1.154 m, a 24.7% reduction, while lowering the P90 error from 2.643 m to 1.612 m.

Index Terms—Indoor positioning, KalmanNet, learned state-space estimation, sensor fusion, pedestrian dead reckoning, Wi-Fi fingerprinting, magnetic localization.

I. INTRODUCTION

Indoor localization supports location-based services, smart buildings, and emergency navigation, where satellite-based positioning is often unreliable indoors [1]. Wi-Fi fingerprinting provides infrastructure-based absolute position estimates [2], pedestrian dead reckoning (PDR) provides high-rate relative motion but accumulates drift [3], and indoor magnetic structure provides an additional spatial fingerprint [4]. These modalities have already been combined in prior systems: Li *et al.* integrate PDR, Wi-Fi fingerprinting, and magnetic matching [5]; Bolat and Akcakoca combine Wi-Fi, magnetic-field, and inertial navigation [6]; and DeepPositioning learns joint Wi-Fi–magnetic localization [7]. Thus, combining the three sensing sources is not itself our contribution. The problem addressed here is how to apply spatial corrections from asynchronous learned measurements to a drifting motion state when measurement availability and reliability change over time.

For a planar state, PDR produces a causal prediction $\mathbf{x}_t^- = \mathbf{x}_{t-1} + \mathbf{u}_t$. A spatial measurement $\mathbf{z}_{j,t}$ from modality $j \in \{wifi, mag\}$ then produces an innovation $\mathbf{y}_{j,t} = \mathbf{z}_{j,t} - \mathbf{x}_t^-$. Fusion therefore requires deciding how strongly each correction $\mathbf{K}_{j,t}\mathbf{y}_{j,t}$ should modify the state. Prior work addresses parts of this problem using adaptive factor-graph weighting for Wi-Fi/PDR [8], neural Wi-Fi/PDR fusion [3], and learned Kalman-style state estimation [9]. The distinction sought here is architectural rather than an initialization fix: two learned spatial measurements are represented in one Cartesian state space and are allowed to influence the causal state update through separate, history-dependent gain matrices.

Learned spatial models provide complementary measurement functions. DeepPositioning combines Wi-Fi and magnetic fingerprints using deep learning [7]; MINLOC applies a CNN to magnetic-field patterns [10]; and Zardkoohi *et al.* use a Bi-LSTM for magnetic-sequence localization [11]. A bidirectional recurrent estimator uses both past and future sequence context and therefore is not directly usable for streaming causal inference without reformulation. Our objective is not to claim that these models fail because of poor initialization, but to separate spatial measurement extraction from temporal correction and to make the contribution of each available measurement explicit in the state update.

In this work, we propose a decoupled causal learned state-space architecture for multimodal indoor localization, where *multimodal* denotes heterogeneous Wi-Fi, magnetic-field, and inertial measurements. First, a Wi-Fi heatmap MLP and gravity-referenced magnetic CNN independently produce Cartesian position measurements; the magnetic CNN also provides a scalar relative-uncertainty score. Second, DualKalmanNet forms separate Cartesian innovations and predicts independent 2×2 Wi-Fi and magnetic gain matrices, while availability masks remove unavailable modalities. Third, a training-normalized magnetic confidence weight explicitly attenuates uncertain magnetic corrections. The temporal fusion experiment evaluates independent map-constrained trajectories within one surveyed environment; device generalization is evaluated separately for the Wi-Fi and static-fingerprint experiments. The architecture can be reused after site-specific surveying and training, but the present fusion results do not constitute building-agnostic or plug-and-play unseen-device validation.

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Section II defines measurements and preprocessing; Section III presents the estimator; Section IV gives training and evaluation; Section V reports results; and Section VI concludes the paper.

II. SENSOR MEASUREMENTS AND PREPROCESSING

Assume a targeted indoor environment with N identifiable Wi-Fi Access Points (APs) and a Cartesian reference grid containing M spatial cells. At time t , the phone-side observations relevant to this work are a Wi-Fi RSSI vector $\mathbf{s}_t \in \mathbb{R}^N$, a magnetometer vector $\mathbf{m}_t \in \mathbb{R}^3$, and an accelerometer vector $\mathbf{a}_t \in \mathbb{R}^3$. The PDR module also receives a causal heading observation $\hat{\theta}_t$. The objective is to estimate the planar Cartesian state $\mathbf{x}_t = [x_t, y_t]^T \in \mathbb{R}^2$. Missing Wi-Fi or magnetic measurements are represented explicitly by modality-availability masks during fusion.

a) *Wi-Fi preprocessing.*: The raw Wi-Fi scan $\mathbf{s}_t \in \mathbb{R}^N$ is preprocessed into $\tilde{\mathbf{s}}_t \in [0, 1]^N$. Undetected APs are stored at the -100 dBm floor; clipping and affine rescaling then give

$$\tilde{s}_{t,i} = \frac{\text{clip}(s_{t,i}, -90, -30) + 90}{60}, \quad (1)$$

so an undetected AP maps to zero without a separate branch.

b) *Magnetic preprocessing.*: The magnetic path begins with gravity-referenced survey preprocessing. For each raw static sample, let $\mathbf{m}_t \in \mathbb{R}^3$ be the magnetometer vector and $\mathbf{a}_t \in \mathbb{R}^3$ the accelerometer vector. The implementation uses the instantaneous normalized acceleration

$$\hat{\mathbf{a}}_t = \frac{\mathbf{a}_t}{\|\mathbf{a}_t\|} \quad (2)$$

as a gravity-direction proxy for the static survey record, and computes

$$\begin{aligned} m_N &= \|\mathbf{m}_t\|, & m_V &= \mathbf{m}_t^T \hat{\mathbf{a}}_t, \\ m_H &= \sqrt{\max(m_N^2 - m_V^2, 0)}, & \delta &= \text{atan2}(m_V, m_H). \end{aligned} \quad (3)$$

These quantities are invariant to a common rigid rotation of the phone when $\hat{\mathbf{a}}_t$ is a valid gravity proxy. The processed survey database stores the mean and standard deviation of each quantity over a static visit. The magnetic-map trainer then subtracts each phone's mean feature value, averages the centered visit means at each surveyed node, and interpolates the four channels (m_N, m_V, m_H, δ) onto the same 1 m Cartesian environment grid used by the Wi-Fi model.

The per-phone centering defines a survey-specific magnetic feature domain. The present fusion evaluation operates inside that domain and does not establish that an uncalibrated unseen handset can enter it without an additional causal normalization procedure.

c) *Inertial inputs.*: For PDR, the accelerometer stream is reduced to its magnitude $a_t = \|\mathbf{a}_t\|$. The heading observation $\hat{\theta}_t$ is treated as a causal external input to the estimator rather than as a quantity inferred from future trajectory information. The present synthetic evaluation generates this heading observation from the latent path plus a specified noise process

in Section IV; a deployable raw-sensor heading estimator is outside the scope of the current experiment.

III. PROPOSED STATE-SPACE ESTIMATOR

The estimator forms Wi-Fi and magnetic Cartesian measurements, propagates PDR motion, and combines them with DualKalmanNet (Fig. 1).

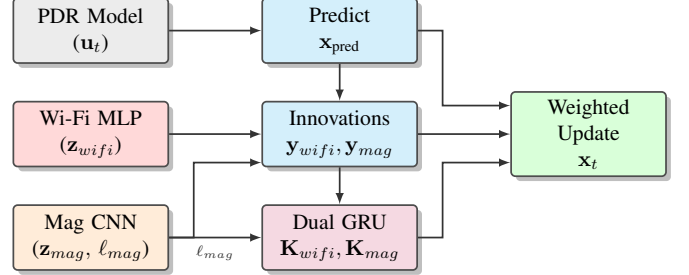


Fig. 1: The proposed Neural-Kalman fusion architecture. The Wi-Fi heatmap and magnetic CNN produce Cartesian innovations against the PDR prediction. The magnetic CNN also provides a learned scalar log-uncertainty score ℓ_{mag} . A recurrent Dual GRU predicts independent 2×2 gain matrices, while relative magnetic uncertainty explicitly weights the magnetic correction before the posterior update.

A. Wi-Fi Heatmap Measurement Model

To map this signal to the surveyed grid, the MLP processes $\tilde{\mathbf{s}}_t$ and outputs a Softmax heatmap $\mathbf{p} \in [0, 1]^M$ satisfying

$$p_c \geq 0, \quad \sum_{c=1}^M p_c = 1. \quad (4)$$

Here p_c is the learned discrete probability mass assigned to grid cell c .

At inference, the Cartesian Wi-Fi measurement used by the final fusion network is the Softmax expectation

$$\mathbf{z}_{wifi} = \sum_{c=1}^M p_c \mathbf{c}_c, \quad (5)$$

where $\mathbf{c}_c \in \mathbb{R}^2$ is the physical coordinate of grid cell c . The final DualKalmanNet consumes $\mathbf{z}_{wifi}$ directly; a heatmap covariance is not part of its 13-dimensional GRU input.

B. Magnetic Sequence Measurement Model

The magnetic model consumes the four preprocessed channels over a causal window ending at time t :

$$\mathbf{M}_t \in \mathbb{R}^{T \times 4}, \quad T = 84, \quad (6)$$

No future frame is included. The CNN maps $\mathbf{M}_t$ to $\mathbf{z}_{mag}$ and ℓ_{mag} .

For $T = 84$, the encoder maps $[B, 4, 84]$ to $[B, 32, 84]$, $[B, 32, 42]$, $[B, 64, 42]$, $[B, 64, 21]$, and finally $[B, 128, 21]$. Adaptive average pooling aggregates all 21 temporal locations into a shared 128-dimensional vector. The position head is $128 \rightarrow 64 \rightarrow 2$ with ReLU and dropout $p = 0.2$, while the

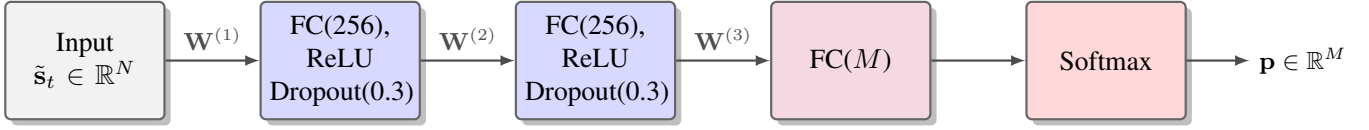


Fig. 2: Architecture of the Wi-Fi Multi-Layer Perceptron (MLP) environment model. Dense layers sequentially extract features from the RSSI vector to compute an M -dimensional probability heatmap.

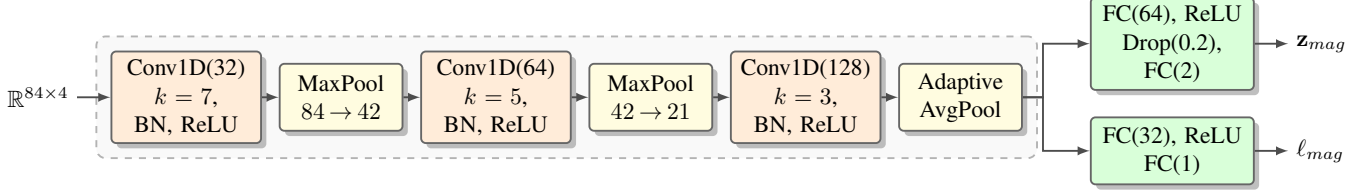


Fig. 3: Code-accurate architecture of the magnetic sequence CNN for $T = 84$. The shared Conv1D encoder reduces the temporal length $84 \rightarrow 42 \rightarrow 21$ before adaptive global averaging into a 128-dimensional representation. Separate heads predict the 2D magnetic position fix $\mathbf{z}_{mag}$ and a scalar log-uncertainty score ℓ_{mag} .

scalar uncertainty head is $128 \rightarrow 32 \rightarrow 1$ with ReLU in its hidden layer. Thus the second output is learned from the same complete 84-frame representation as the position estimate; it is not computed analytically from the position residual at inference time.

ℓ_{mag} is used only as a relative uncertainty indicator; its training objective is given in Section IV.

C. Causal PDR Motion Model

PDR propagates relative motion between absolute updates [3], [12]. From causal acceleration magnitude a_t and heading $\hat{\theta}_t$, it detects steps and forms planar controls. An exponential moving average (EMA) tracks the gravitational baseline,

$$\bar{a}_t = \alpha \bar{a}_{t-1} + (1 - \alpha) a_t, \quad \alpha = 0.98, \quad (7)$$

with $\bar{a}_0 = 9.81 \text{ m/s}^2$. The high-pass residual is

$$\tilde{a}_t = a_t - \bar{a}_t, \quad (8)$$

and a step is declared when

$$\tilde{a}_t > \tau \quad \text{and} \quad (t - t_{\text{last}}) > \Delta_r, \quad (9)$$

where $\tau = 0.6 \text{ m/s}^2$ and $\Delta_r = \lfloor 0.3 f_s \rfloor = 5$ frames. This detector is causal: its state at time t depends only on acceleration samples observed up to t .

For a detected step, the PDR control is

$$\mathbf{u}_t = \begin{bmatrix} L_s \cos \hat{\theta}_t \\ L_s \sin \hat{\theta}_t \end{bmatrix}, \quad (10)$$

where $L_s = 0.65 \text{ m}$ is fixed in the evaluated estimator and $\hat{\theta}_t$ is an external causal heading input; its synthetic noise model is specified in Section IV. No ground-truth path length is used to adapt L_s .

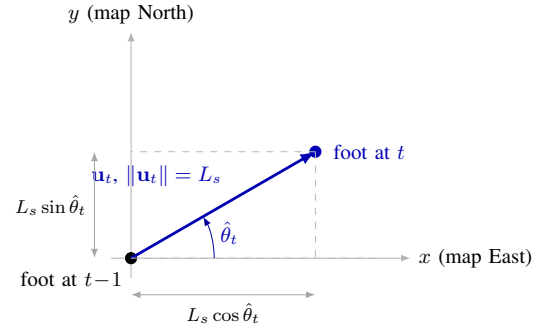


Fig. 4: Geometry of the PDR control used by the estimator. A detected step of nominal length L_s is projected along the currently available heading observation $\hat{\theta}_t$. In the synthetic fusion evaluation, $\hat{\theta}_t$ is a noisy simulated heading measurement rather than a ground-truth-calibrated `Orn_z` signal.

D. Dual-Innovation KalmanNet Fusion

For an Extended Kalman Filter (EKF), after linearizing the measurement model with Jacobian $\mathbf{H}_t$, the analytical correction gain is

$$\mathbf{K}_t = \mathbf{P}_t^- \mathbf{H}_t^T (\mathbf{H}_t \mathbf{P}_t^- \mathbf{H}_t^T + \mathbf{R}_t)^{-1}, \quad (11)$$

where $\mathbf{P}_t^-$ is the predicted state-error covariance and $\mathbf{R}_t$ is the assumed measurement-noise covariance. KalmanNet preserves a state-space prediction/correction structure but replaces this analytical gain computation with a recurrent learned mapping when parts of the dynamics or statistics are imperfectly known [9]. Our dual extension predicts separate gains for Wi-Fi and magnetic measurements. The causal PDR prediction is

$$\mathbf{x}_{\text{pred}} = \mathbf{x}_{t-1} + \mathbf{u}_t. \quad (12)$$

Both learned environment models are then expressed in the same Cartesian state space. The Wi-Fi innovation is

$$\mathbf{y}_{wif i} = \mathbf{z}_{wif i} - \mathbf{x}_{\text{pred}} \in \mathbb{R}^2, \quad (13)$$

and, crucially, the magnetic innovation directly uses the 2D output of the magnetic CNN:

$$\mathbf{y}_{mag} = \mathbf{z}_{mag} - \mathbf{x}_{pred} \in \mathbb{R}^2. \quad (14)$$

The pair $(\mathbf{y}_{wif}, \mathbf{y}_{mag})$ constitutes the *dual innovation*: two Cartesian measurement residuals formed against the same PDR prediction. The magnetic CNN output $\mathbf{z}_{mag}$ therefore enters the correction in the same state space as the Wi-Fi measurement.

The CNN additionally predicts the scalar log-uncertainty score $\ell_{mag,t}$ and its positive scale $q_{mag,t} = \exp(\ell_{mag,t})$. Because only relative reliability is used in fusion, we normalize the score using a training-only reference

$$\begin{aligned} \ell_{ref} &= \text{median}_{(w,t) \in \mathcal{D}_{train}: m_{mag}=1} \ell_{mag,w,t}, \\ q_{ref} &= \exp(\ell_{ref}). \end{aligned} \quad (15)$$

The relative magnetic confidence weight is then

$$\begin{aligned} w_{mag,t} &= \frac{1}{1 + \exp(\ell_{mag,t} - \ell_{ref})} \\ &= \frac{1}{1 + q_{mag,t}/q_{ref}}, \quad 0 < w_{mag,t} < 1. \end{aligned} \quad (16)$$

A magnetic prediction with an uncertainty score above the training reference is therefore down-weighted, whereas a relatively confident fix retains a larger correction. The log-uncertainty score itself is also retained as a GRU input so that the network may learn additional context-dependent reliability beyond this explicit weighting.

For readability, the 13-dimensional GRU input is listed explicitly below. Define the previous posterior displacement and the masked, numerically clipped magnetic-confidence feature as

$$\begin{aligned} \Delta \mathbf{x}_{t-1} &= \mathbf{x}_{t-1} - \mathbf{x}_{t-2}, \\ c_{mag,t} &= m_{mag} \text{ clip}(\ell_{mag,t}, -6, 8). \end{aligned} \quad (17)$$

Then the input groups are

GRU feature	Dim.
Wi-Fi innovation $\mathbf{y}_{wif}$	2
Magnetic innovation $\mathbf{y}_{mag}$	2
Wi-Fi fix difference $\Delta \mathbf{z}_{wif}$	2
PDR control $\mathbf{u}_t$	2
Previous posterior displacement $\Delta \mathbf{x}_{t-1}$	2
Wi-Fi availability m_{wif}	1
Magnetic availability m_{mag}	1
Magnetic confidence $c_{mag,t}$	1
Total	13

Here,

$$\Delta \mathbf{z}_{wif,t} = \mathbf{z}_{wif,t} - \mathbf{z}_{wif,t-}, \quad (18)$$

where t^- is the most recent time with an available Wi-Fi fix; the feature is zero when no new Wi-Fi measurement is present. It provides a short-term Wi-Fi consistency cue that the GRU interprets jointly with the current innovation, PDR displacement, and recurrent history. The GRU outputs eight values, reshaped into independent 2×2 matrices $\mathbf{K}_{wif}$ and $\mathbf{K}_{mag}$. The posterior update is

$$\mathbf{x}_t = \mathbf{x}_{pred} + m_{wif} \mathbf{K}_{wif} \mathbf{y}_{wif} + m_{mag} w_{mag,t} \mathbf{K}_{mag} \mathbf{y}_{mag}. \quad (19)$$

The availability masks preserve operation under missing modalities: setting $m_{wif} = 0$ or $m_{mag} = 0$ removes the corresponding correction. When a magnetic window is available but its predicted uncertainty score is large, $w_{mag,t}$ suppresses that correction before it can dominate the state estimate. This combines the learned, context-dependent gains of KalmanNet with an explicit relative-confidence prior from the magnetic CNN.

IV. EXPERIMENTAL SETUP AND TRAINING

We evaluate the proposed fusion architecture in the IT Engineering building of the MagWi Benchmark Dataset [13]. The surveyed fingerprints define the environment model used to generate Wi-Fi and magnetic measurements, while temporal fusion is evaluated on newly generated map-constrained trajectories. Consequently, the fusion experiment tests generalization to unseen trajectories within a fixed surveyed environment; it is not, by itself, a held-out-smartphone experiment.

A. Measurement-Model Training Objectives

The Wi-Fi and magnetic measurement models are trained separately from temporal fusion.

a) *Wi-Fi heatmap.*: The training target $\mathbf{q} \in \mathbb{R}^M$ is a discrete probability distribution constructed by evaluating a 2D isotropic Gaussian, centered at the true position $\mathbf{x}_{true}$, at each of the M grid cell coordinates $\mathbf{c}_c$, and normalizing the result to sum to unity:

$$q_c = \frac{\exp\left(-\frac{\|\mathbf{c}_c - \mathbf{x}_{true}\|^2}{2\sigma^2}\right)}{\sum_{j=1}^M \exp\left(-\frac{\|\mathbf{c}_j - \mathbf{x}_{true}\|^2}{2\sigma^2}\right)}, \quad \sum_{c=1}^M q_c = 1 \quad (20)$$

where σ is the spatial spread parameter. The Gaussian is used only to construct the discrete training target $\mathbf{q}$; the network output $\mathbf{p}$ remains an unconstrained learned Softmax distribution over cells. Relative to a one-hot target, $\mathbf{q}$ assigns nonzero mass to nearby cells and therefore provides spatially graded supervision. The network is optimized by minimizing the KL divergence between $\mathbf{q}$ and $\mathbf{p}$:

$$\mathcal{L}_{MLP} = D_{KL}(\mathbf{q} \parallel \mathbf{p}) = \sum_{c=1}^M q_c \log\left(\frac{q_c}{p_c}\right) \quad (21)$$

b) *Magnetic sequence CNN.*: The CNN is trained on survey-derived, map-constrained magnetic sequences rather than directly on the raw MagWi continuous recordings. Random corridor paths are sampled from the surveyed node graph, interpolated at 16.7 Hz, and used to bilinearly sample the four-channel magnetic map; channel-wise Gaussian noise is added using the measured within-node variability.

Each generated 84-frame window uses the Cartesian position at its final frame as the regression target.

Let $\ell_{mag}^{(i)}$ denote the scalar score for training window i and define $q_{mag}^{(i)} = \exp(\ell_{mag}^{(i)})$. The historical training implementation uses the uncertainty-weighted regression objective

$$\mathcal{L}_{CNN} = \frac{1}{B} \sum_{i=1}^B \left[\frac{\|\mathbf{z}_{mag}^{(i)} - \mathbf{z}_{true}^{(i)}\|^2}{2\tilde{q}_{mag}^{(i)}} + \frac{1}{2}\ell_{mag}^{(i)} \right], \quad (22)$$

where $\tilde{q}_{mag}^{(i)} = \max(q_{mag}^{(i)}, 0.01)$. A difficult sample can therefore receive a larger learned scale, reducing the first term, while the positive log-scale penalty discourages arbitrarily large uncertainty scores.

Importantly, (22) is *not* the exact negative log-likelihood of a two-dimensional isotropic Gaussian: for a covariance $q\mathbf{I}_2$, the Gaussian normalization contributes $\log q$, rather than $\frac{1}{2}\log q$. We therefore do not interpret q_{mag} as a calibrated Cartesian variance or covariance. It is used only as a learned relative uncertainty/difficulty scale. The separate calibration analysis shows that this score has useful reliability ordering but a conservative absolute scale; the fusion stage consequently uses only its training-normalized relative value.

B. Survey-Derived Synthetic Trajectory Evaluation

The final fusion benchmark is synthetic because the continuous MagWi recordings do not provide the per-frame trajectory ground truth required for supervised KalmanNet training and controlled temporal evaluation. The simulator has access to the generated ground-truth state in order to produce sensor observations; the estimator does not. Specifically:

- **Survey environment:** An ε -graph ($\varepsilon = 1.6$ m) is built from surveyed IT Engineering reference-node coordinates. The processed survey database is also used to construct a pool of real Wi-Fi scans and an interpolated four-channel magnetic map. These environment resources include the available surveyed devices; therefore the fusion results are not claimed as an unseen-device test.
- **Trajectory generation:** Random endpoint pairs are sampled on the connected corridor graph and joined by shortest paths. Each path is interpolated at 16.7 Hz with walking speed sampled uniformly from 1.0 to 1.35 m/s. Fusion-training and fusion-test walks are generated independently with fixed seeds 1 and 2, respectively. The implementation hashes the resulting binned target trajectories and aborts if an identical trajectory occurs in both sets.
- **PDR measurements:** The simulator computes the geometric path tangent θ_t^{true} and forms a noisy heading observation

$$\hat{\theta}_t = \theta_t^{true} + b_t + \epsilon_t, \quad (23)$$

where b_t is a Gaussian random walk with per-frame standard deviation $0.5^\circ/\sqrt{16.7}$ and ϵ_t is zero-mean white noise with standard deviation 8.8° . Acceleration magnitude is synthesized as a step-frequency sinusoid (frequency sampled from 1.7–2.0 Hz) around gravity with additive Gaussian noise, then passed through the causal step detector of Section III-C. Every detected step uses the fixed nominal $L_s = 0.65$ m.

- **Wi-Fi measurements:** At each scheduled scan time, the nearest surveyed node to the true simulated position is located, one stored RSSI fingerprint from that node is sampled, and the actual trained Wi-Fi heatmap network produces $\mathbf{z}_{wifi}$. The degraded regime changes the scan interval to 5 s and independently drops 40% of AP entries before inference.
- **Magnetic measurements:** The four rotation-invariant magnetic channels are bilinearly sampled from the survey-derived magnetic map along the simulated path, perturbed using the measured map-noise scale, and passed through the actual 84-frame magnetic CNN. Only causal windows ending at the current fusion time produce $\mathbf{z}_{mag}$ and ℓ_{mag} .
- **Estimator/ground-truth separation:** KalmanNet receives only binned PDR controls, Wi-Fi fixes and masks, magnetic CNN fixes and masks, and magnetic log-uncertainty score. The generated ground-truth positions are used only to construct the simulated sensor observations and as the supervised training/evaluation target; they are never included in the KalmanNet input vector.

We generate 250 fusion-training trajectories and 60 independent fusion-test trajectories, each represented by 160 fusion bins. This protocol provides controlled temporal ground truth while preserving a strict separation between the simulator's latent state and the observations supplied to the estimator.

C. Training Details

All neural network components were developed using the PyTorch framework. The MLP and CNN were optimized via Adam ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}) with an initial learning rate of 10^{-3} , dynamically decayed via a ReduceLROnPlateau scheduler (patience 8, factor 0.5). The MLP uses a Dropout rate of $p = 0.3$ and a Gaussian target with spatial standard deviation $\sigma = 2.0$ m. The MLP was trained for 80 epochs; the CNN for 60 epochs. The 1D-CNN magnetic sequence matcher uses $T = 84$ frames (5.0 s at 16.7 Hz), selected from the candidate set $\{50, 84, 134, 167\}$ by the development sweep. The KalmanNet GRU was trained for 150 epochs via Adam with a learning rate of 2×10^{-3} , weight decay of 10^{-5} , and MSE loss against ground-truth trajectories. For each signal-availability regime, the reference magnetic log-uncertainty score ℓ_{ref} in (15) is computed only from the 250 fusion-training trajectories and is then frozen for evaluation.

Device generalization is evaluated separately from temporal fusion. In the standalone Wi-Fi heatmap phone-split experiment and the real static KNN baseline, Samsung Galaxy S9+ fingerprints are held out from model fitting and used only for evaluation. The fusion benchmark instead uses survey-derived environment resources and evaluates 60 unseen synthetic trajectories; its reported KalmanNet errors should therefore be interpreted as trajectory-generalization results within the surveyed IT Engineering environment, not as held-out-device performance.

V. EXPERIMENTAL RESULTS

A. Environment Model Accuracy

The standalone performance of the Wi-Fi heatmap environment model prior to temporal fusion is shown in Table I. The mixed-device random split gives 1.43 m MAE, while the separate phone-split experiment gives 2.02 m MAE when Samsung Galaxy S9+ fingerprints are held out from model fitting.

TABLE I: Wi-Fi Heatmap Environment Model Performance

Configuration	Test Device	MAE	Med.	P90	Max
Random split	Mixed	1.43 m	1.12 m	2.65 m	7.84 m
Held-out device	Samsung S9+	2.02 m	1.68 m	3.74 m	9.12 m

For a classical non-neural reference, we additionally evaluate K-nearest-neighbor (KNN) localization on the MagWi fingerprints [13]. Figure 5(a) reports a real held-out-device experiment: 726 fingerprint visits from A8, G7, and S8 are used for training and 90 S9+ visits are reserved for final testing. Hyperparameters are selected only from the training phones using leave-one-phone-out cross-validation. Wi-Fi-only KNN achieves 3.31 m mean error (1.57 m median), whereas magnetic-only KNN reaches 17.54 m and a simple Wi-Fi-magnetic feature concatenation reaches 7.46 m. The degradation of the naive magnetic variants highlights the strong cross-device sensitivity of static magnetic fingerprints and motivates learned, sequence-based magnetic matching rather than direct raw-feature concatenation.

B. Dual-Innovation KalmanNet Fusion Comparison

Table II details the comparative performance of the temporal fusion mechanisms across the independent test walks, evaluated under two separate signal availability regimes.

TABLE II: Fusion Comparison on the Matched 60-Walk Protocol

Regime	Model	Mean Err.	Med.
Full Wi-Fi (1 Hz)	Wi-Fi-only KalmanNet	0.473 ± 0.035 m	0.449 m
	Wi-Fi+Mag KNN (non-temp.)	0.802 ± 0.038 m	0.772 m
	CNN DualKalmanNet	0.506 ± 0.056 m	0.440 m
	CNN Dual + rel. confidence	0.494 ± 0.046 m	0.437 m
Degraded Wi-Fi (5 s, 40% drop)	Wi-Fi-only KalmanNet	1.533 ± 0.193 m	1.392 m
	Wi-Fi+Mag KNN (non-temp.)	2.606 ± 0.184 m	2.560 m
	CNN DualKalmanNet	1.171 ± 0.139 m	1.042 m
	CNN Dual + rel. confidence	1.154 ± 0.129 m	1.113 m

For the matched trajectory comparison in Fig. 5(b)–(c), KNN is deliberately non-temporal, with K selected by five-fold grouped cross-validation that keeps each trajectory within one fold. Under full 1 Hz Wi-Fi it obtains 0.802 m mean error,

versus 0.473 m for Wi-Fi-only KalmanNet and 0.494 m for uncertainty-weighted CNN fusion; the latter also improves the median from 0.449 m to 0.437 m. Under degraded Wi-Fi, KNN reaches 2.606 m, Wi-Fi-only KalmanNet 1.533 m, unweighted CNN DualKalmanNet 1.171 m, and the uncertainty-weighted model 1.154 m. Thus relative magnetic uncertainty gives a 24.7% mean-error reduction over Wi-Fi-only KalmanNet and lowers the unweighted model’s P90 from 2.064 m to 1.612 m. The KNN gap shows that PDR propagation and recurrent context matter beyond simply having both absolute estimates available.

Figure 6 shows the same degraded regime geometrically. Open-loop PDR drifts through corridor turns and Wi-Fi-only KalmanNet can re-anchor only at sparse scans, whereas the CNN supplies causal magnetic position innovations between scans and relative uncertainty suppresses unreliable corrections. The fused path therefore remains closer to the surveyed route. The displayed walk is a central-performance case rather than the best improvement; its exact selection data are stored with the benchmark output.

VI. CONCLUSION

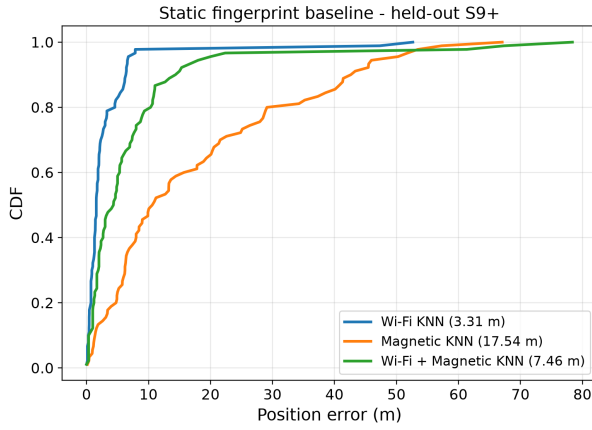
This paper presented a decoupled learned state-space architecture that separates spatial inference from temporal tracking. A Wi-Fi heatmap MLP and gravity-referenced magnetic sequence CNN provide Cartesian measurements, while DualKalmanNet predicts independent 2×2 gains; availability masks and relative magnetic uncertainty suppress missing or unreliable corrections. Under degraded Wi-Fi, the uncertainty-weighted model reduces mean error from 1.533 m for Wi-Fi-only KalmanNet to 1.154 m (24.7%) and reduces the high-error tail. The present fusion results demonstrate trajectory generalization within a surveyed environment and still require building-specific environment resources. Future work will extend the state to three-dimensional, multi-floor positioning, including barometric altitude information.

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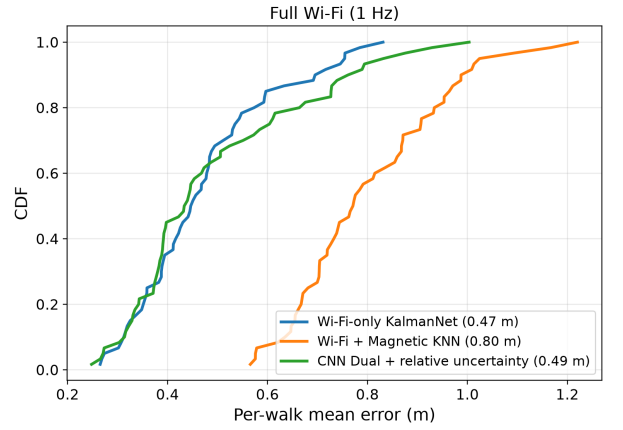
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REFERENCES

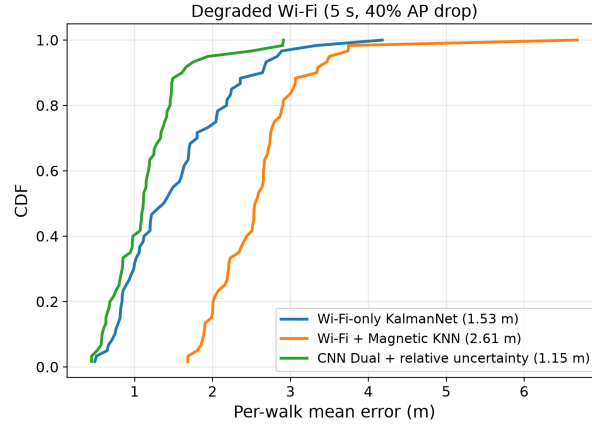
- [1] P. Bahl and V. N. Padmanabhan, “Radar: An in-building rf-based user location and tracking system,” in *Proc. IEEE INFOCOM*, vol. 2, pp. 775–784, 2000.
- [2] M. Youssef and A. Agrawala, “The horus wlan location determination system,” in *Proc. ACM MobiSys*, pp. 205–218, 2005.
- [3] D. Yu, C. Li, and J. Xiao, “Neural networks-based wi-fi/pdr indoor navigation fusion methods,” *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–14, 2023.
- [4] I. Ashraf, S. Din, S. Hur, G. Kim, and Y. Park, “Empirical overview of benchmark datasets for geomagnetic field-based indoor positioning,” *Sensors*, vol. 21, no. 10, p. 3533, 2021.
- [5] Y. Li, Y. Zhuang, H. Lan, Q. Zhou, X. Niu, and N. El-Sheimy, “A hybrid wifi/magnetic matching/pdr approach for indoor navigation with smartphone sensors,” *IEEE Communications Letters*, vol. 20, no. 1, pp. 169–172, 2016.



(a) Real static fingerprints with S9+ held out.



(b) Matched trajectory protocol, full Wi-Fi (1 Hz).



(c) Matched trajectory protocol, degraded Wi-Fi.

Fig. 5: KNN baselines and fusion CDFs. Panel (a) is a separate real static-fingerprint held-out-device experiment and is not numerically comparable to the synthetic-walk results in (b)–(c). Panels (b)–(c) use the same 250-training/60-test trajectory protocol as KalmanNet; the non-temporal KNN uses contemporaneous Wi-Fi and magnetic fixes, magnetic log-uncertainty, and availability masks, without PDR or recurrent history.

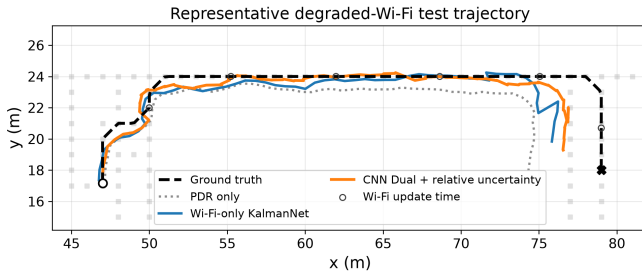


Fig. 6: Representative degraded-Wi-Fi synthetic/map-constrained test walk, selected from the interquartile range of fusion improvements and then by maximum heading-change score. Curves compare ground truth, open-loop PDR, Wi-Fi-only KalmanNet, and the final uncertainty-weighted CNN DualKalmanNet; circles mark Wi-Fi updates.

- [6] U. Bolat and M. Akcakoca, “A hybrid indoor positioning solution based on wi-fi, magnetic field, and inertial navigation,” in *Proc. 14th Workshop on Positioning, Navigation and Communications (WPNC)*, pp. 1–6, 2017.
- [7] W. Zhang, R. Sengupta, J. Fodero, and X. Li, “Deeppositioning: In-

telligent fusion of pervasive magnetic field and wifi fingerprinting for smartphone indoor localization via deep learning,” in *Proc. IEEE Int. Conf. Machine Learning and Applications (ICMLA)*, pp. 7–13, 2017.

- [8] Y. Wang, Z. Li, J. Gao, and L. Zhao, “Deep neural network-based wi-fi/pedestrian dead reckoning indoor positioning system using adaptive robust factor graph model,” *IET Radar, Sonar & Navigation*, vol. 14, no. 1, pp. 36–47, 2020.
- [9] G. Revach, N. Shlezinger, X. Ni, A. L. Escoriza, R. J. G. van Sloun, and Y. C. Eldar, “Kalmannet: Neural network aided kalman filtering for partially known dynamics,” *IEEE Trans. Signal Process.*, vol. 70, pp. 1532–1547, 2022.
- [10] I. Ashraf, M. Kang, S. Hur, and Y. Park, “Minloc: Magnetic field patterns-based indoor localization using convolutional neural networks,” *IEEE Access*, vol. 8, pp. 66213–66227, 2020.
- [11] M. Zardkoohi, Y. Seifi Kavian, and K. Ansari-Asl, “Indoor localization using smartphone magnetic sensor data: A bi-lstm neural network approach,” *IEEE Access*, vol. 13, pp. 165795–165809, 2025.
- [12] C. Isaia, L. Yu, W. Cai, and M. P. Michaelides, “Axes mapping and sensor fusion for attitude-unconstrained pedestrian dead reckoning,” *Sensors*, vol. 26, no. 6, p. 1968, 2026.
- [13] I. Ashraf, S. Din, M. U. Ali, S. Hur, Y. B. Zikria, and Y. Park, “Magwi: Benchmark dataset for long term magnetic field and wi-fi data involving heterogeneous smartphones, multiple orientations, spatial diversity and multi-floor buildings,” *IEEE Access*, vol. 9, pp. 77976–77996, 2021.